PHISHING WEBSITE DETECTION USING MACHINE LEARNING

A Smart Approach to Identify Fraudulent Websites Using Machine Learning

Internship Project

INTRODUCTION

- Phishing is a cyber attack used to steal sensitive information.
- Internet usage is growing rapidly .
- Attackers create fake websites that look legitimate
- Cyber attacks and phishing websites are increasing.
- Users often cannot distinguish between genuine and fake websites.
- Users may unknowingly share passwords, banking details, and personal data.
- Machine Learning provides an intelligent solution for website classification.

PROBLEM STATEMENT

- Existing Problem:
- Fake websites steal sensitive information.
- Manual verification is difficult.

- Proposed Solution:
- Automatically predict whether a website is Legitimate or Phishing.

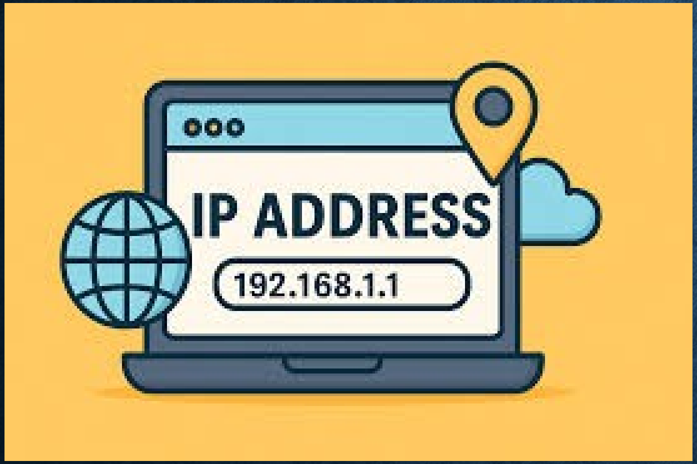
PROJECT OBJECTIVES

- Detect phishing websites accurately.
- Improve user security.
- Provide quick and accurate predictions
- Develop a user-friendly web application.
- Provide real-time URL prediction.

TECHNOLOGIES USED

- Python
- Pandas
- Numpy
- Scikit- Learn
- Joblib
- Streamlit

DATASET OVERVIEW



- Features:
- Number of Dots,URL Length
- Number of Dashes
- Presence of @ Symbol
- IP Address Detection
- HTTPS in Hostname
- Path Level
- Path Length
- Number of Numeric Character

- Target: Phising

METHODOLOGY

- Dataset Collection → Data Preprocessing → Feature Selection → Train-Test Split → Model Training → Performance Evaluation → Website Prediction

ALGORITHMS USED

- Random Forest Classifier

why Random Forest

- High accuracy
- Handles large datasets efficiently
- Reduces overfitting
- Suitable for classification problems

SYSTEM ARCHITECTURE

User URL

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Feature Extraction

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Random Forest Model

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Prediction

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Legitimate / Phishing

STREAMLIT APPLICATION

Streamlit

User Interface

- **User enter a URL**
- **Features are extracted automatically**
- **Model predicts whether URL is safe or phishing**
- **Result displayed instantly**

RESULTS

Sample outputs

- Example 1:

  URL: https://www.google.com

  Result: Legitimate URL

- Example 2:

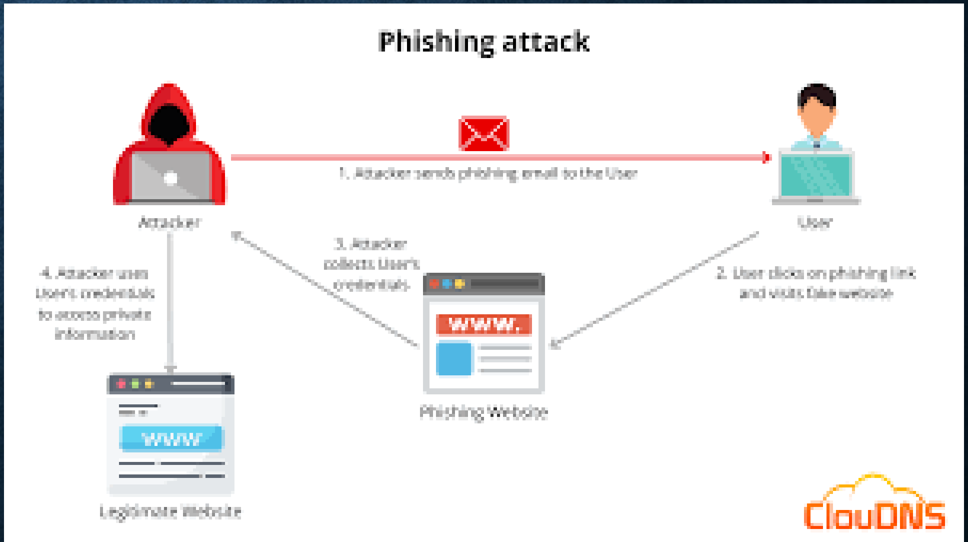
  Suspicious URL

  Result: Phishing URL

LIVE DEMONSTRATION

- User URL → Feature Extraction → ML Model → Prediction

- Output:
- Legitimate Website
- Phishing Website



CONCLUSION & FUTURE SCOPE

- Successfully developed a phishing website detection system.
- Random Forest model effectively classifies URLs.
- Streamlit provides a simple and interactive user interface.
- The system helps users identify suspicious websites quickly.
- Future Scope:
- WHOIS, DNS Verification, SSL Validation, Extension.

THANK YOU